NVDLA on PetaLinux 2024.1

Correctness, latency, throughput, and monitored energy at a glance

100 / 100

LeNet stability

One boot; no module reload or PL reset

246 / 246

ResNet-50 hardware layers

Exact match to source-built nv_small VP golden

1.30x

ResNet-50 execution speedup

507.7 ms NVDLA vs 661.8 ms CPU

-27.7%

ResNet-50 active energy

1.781 J NVDLA vs 2.463 J CPU

-77.6%

ResNet-50 incremental energy

0.134 J NVDLA vs 0.597 J CPU

40

Selected benchmark sessions

Eight balanced cohorts; five fresh boots each

Comparison boundary: deployed-system comparison of nv_small NVDLA INT8 against ONNX Runtime FP32 on four Cortex-A53 cores. Power is the sum of monitored PS and PL rails, not external 12 V board input.

Correctness before performance

Layered acceptance

- OK — Pinned inputs
- OK — ABI and build
- OK — Verified nv_small VP
- OK — Driver probe
- OK — GEM mapping
- OK — IRQ + engine completion
- OK — Exact tensor + repeat

LeNet / MNIST

Fast repeat oracle

INPUT	LOADABLE	HARDWARE LAYERS
1 x 1 x 28 x 28	0.446 MB	10

ENGINE MIX	VP	BOARD
4 Conv / 4 SDP / 2 PDP	Exact output	Exact; 100/100 repeats

Output: 0 2 0 0 0 0 0 124 0 0

ResNet-50

Large multi-engine graph

INPUT	LOADABLE	HARDWARE LAYERS
1 x 3 x 224 x 224	25.77 MB	246

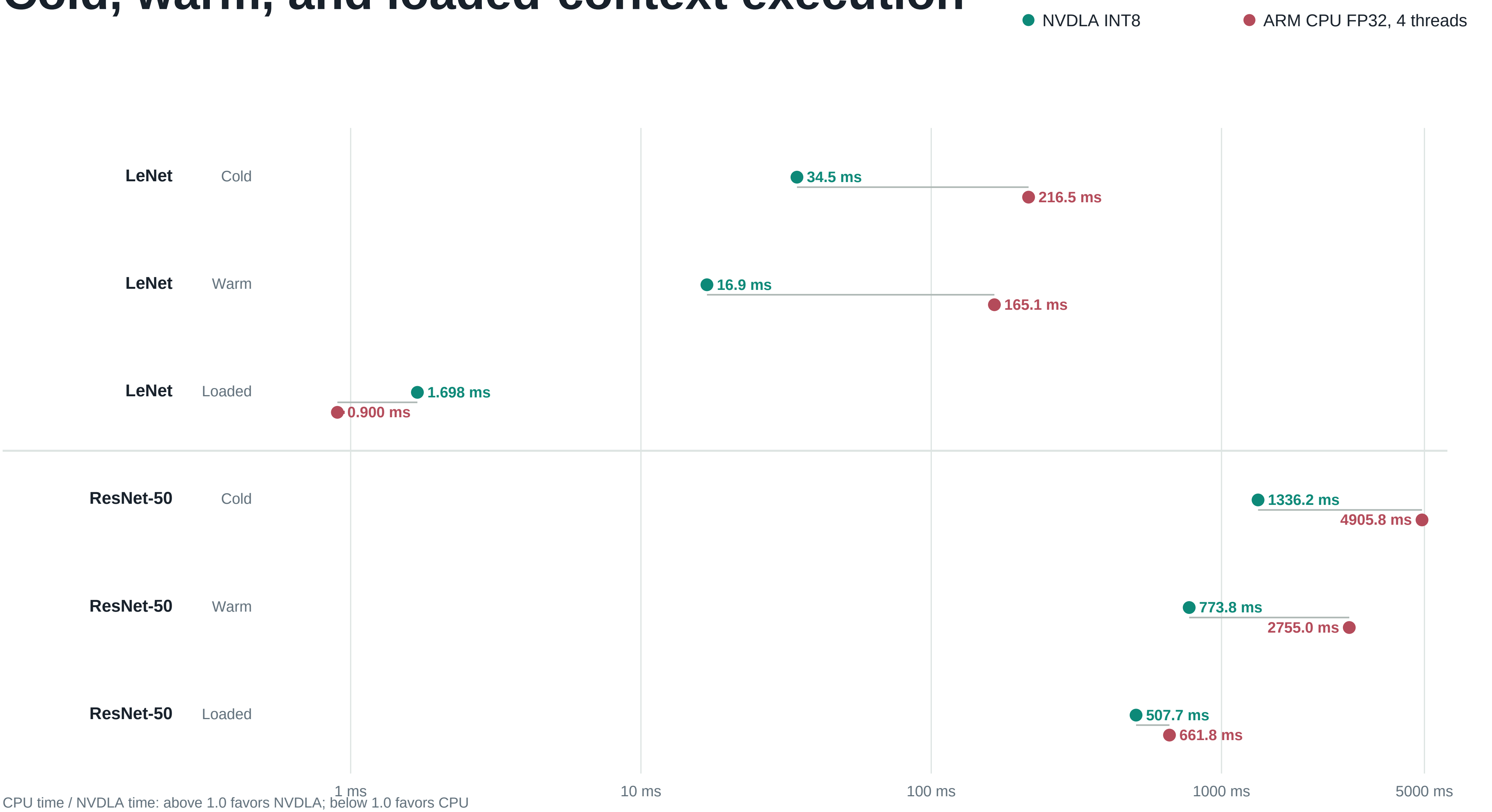
ENGINE MIX	VP	BOARD
114 Conv / 130 SDP / 2 PDP	246/246; golden established	246/246; exact hash

Output: 1,000 signed values; SHA-256 842d34f...

Sources: VP configuration audit, differential trace, board stability and ResNet-50 artifacts

Correctness-qualified samples; no outliers discarded

Cold, warm, and loaded-context execution



CPU time / NVDLA time: above 1.0 favors NVDLA; below 1.0 favors CPU

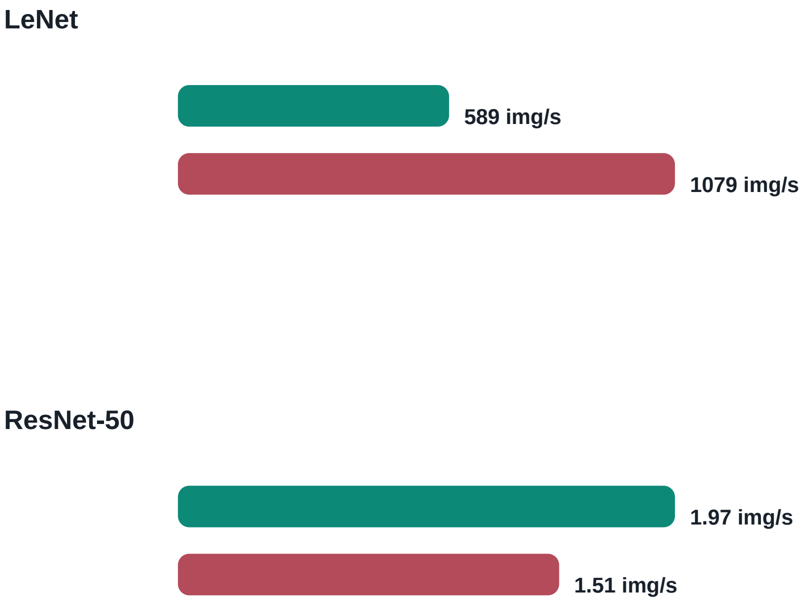
6.28x	9.78x	0.53x	3.67x	3.56x	1.30x
LeNet cold	LeNet warm	LeNet loaded	ResNet cold	ResNet warm	ResNet loaded

What changes between LeNet and ResNet-50?

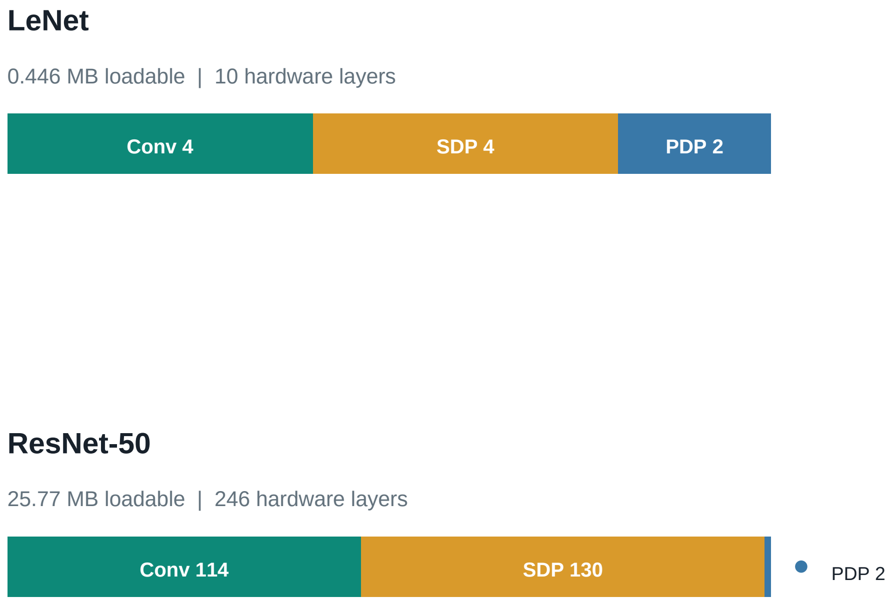
NVDLA INT8

ARM CPU FP32, 4 threads

Loaded-context throughput

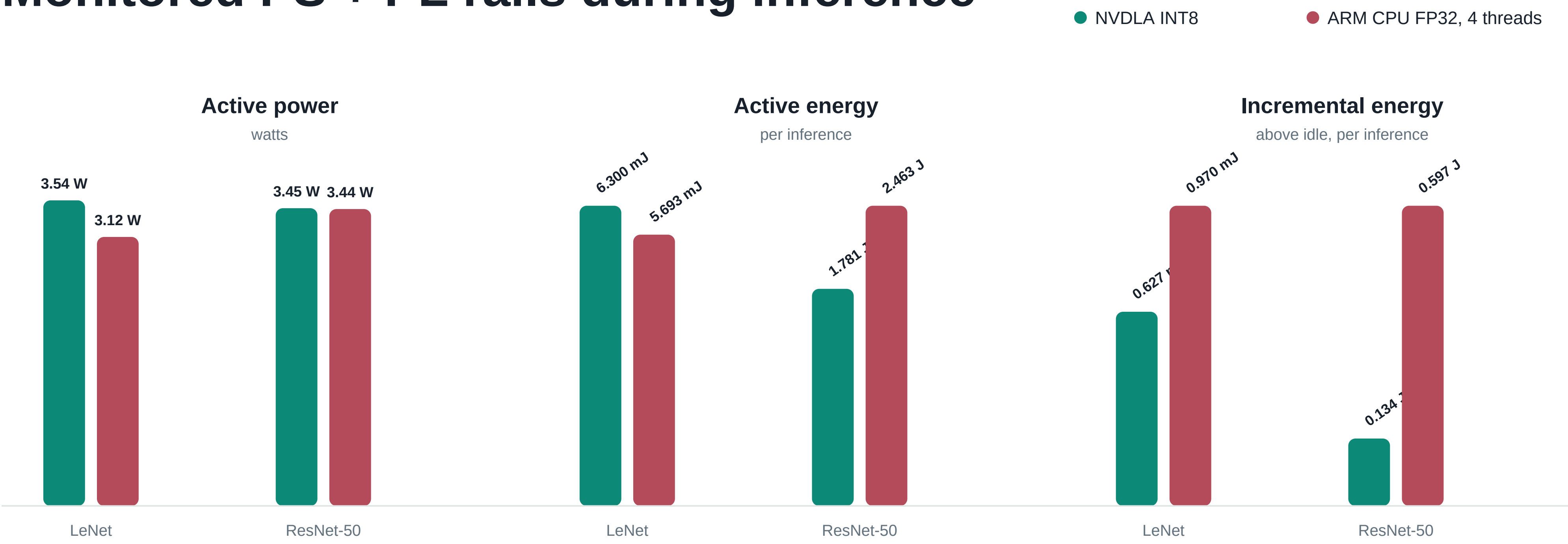


Workload scale and operation mix



ResNet-50 has **24.6x** more hardware layers and a **57.8x** larger loadable. Its larger execution graph amortizes fixed submission and interrupt costs.

Monitored PS + PL rails during inference



LeNet

+10.7% **active energy**

-35.3% incremental

Active includes the complete monitored platform window; incremental subtracts driver-loaded idle.

ResNet-50

-27.7% **active energy**

-77.6% incremental

Active includes the complete monitored platform window; incremental subtracts driver-loaded idle.

What was collected, and how to read it

1. **Fresh boot** — Unique Linux boot ID
2. **Settle + verify** — Clock, frequency, hashes
3. **Correctness gate** — Golden output + kernel health
4. **Measure** — Unpowered latency or powered batch
5. **Archive** — Raw profiles, rails, logs
6. **Aggregate** — Five session medians + 95% CI

Metric inventory

Correctness	Output hashes, exact tensors, engine sequence, IRQ deltas, repeat failures
Latency	Cold deployment, warm deployment, loaded-context execution, runtime phases
Power	18 PS + PL rails, idle/active/incremental watts, integrated energy/inference
Statistics	Raw samples, mean/median, spread, percentiles, retained outliers, bootstrap CI
Workload	Input shape, loadable/model size, HWL count, per-engine operation count
Throughput	Images/s for every timing boundary; CPU/NVDLA latency ratios
Environment	Kernel, binary hashes, clock, CPU affinity/frequency/governor, boot ID
Evidence	Serial, dmesg, runtime profiles, sensor samples, manifests, selection hashes

Interpret with: one ZCU102 and nv_small implementation; NVDLA INT8 versus CPU FP32; monitored rails rather than wall input; five independent boots per final cohort.